

Solving and Analyzing Linear Systems: Worked Examples

High-Level Overview

To determine whether a system of linear equations is consistent, has a unique solution, or has infinitely many solutions, we rely on **Rank Analysis of Augmented Matrices**.

By using elementary row transformations, we reduce the augmented matrix $[A \mid B]$ to an **Echelon Form** (triangular form) and compare:

- The rank of the coefficient matrix $\rho(A)$.
- The rank of the full augmented matrix $\rho(A \mid B)$.
- The total number of unknowns (n).

1. Testing Consistency (Numerical Examples)

Example 1: Testing a 2×2 System

Problem: Test for consistency of the system:

$$x + 5y = 7, \quad -2x - 7y = -5$$

Solution:

- Write the system as a matrix equation $AX = B$:

$$\begin{bmatrix} 1 & 5 \\ -2 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \end{bmatrix}$$

- Form the augmented matrix $[A \mid B]$:

$$[A \mid B] = \left[\begin{array}{cc|c} 1 & 5 & 7 \\ -2 & -7 & -5 \end{array} \right]$$

- Apply row operation $R_2 \rightarrow R_2 + 2R_1$:

$$\sim \left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 3 & 9 \end{array} \right]$$

- Analysis:** Here, $\rho(A) = \rho(A \mid B) = 2$, which matches the number of unknowns ($n = 2$).
- Conclusion:** The system is **consistent and has a unique solution**.

Example 2: Testing a 3×3 System

Problem: Test for consistency of the system:

$$x + y + z = 6, \quad x - y + 2z = 5, \quad 3x + y + z = 8$$

Solution:

- Set up the augmented matrix $[A \mid B]$:

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 3 & 1 & 1 & 8 \end{array} \right]$$

2. Apply row transformations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & -2 & -2 & -10 \end{array} \right]$$

3. Apply $R_3 \rightarrow R_3 - R_2$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & -3 & -9 \end{array} \right]$$

4. **Analysis:** We get $\rho(A) = \rho(A \mid B) = 3$, which equals the number of unknowns ($n = 3$).

5. **Conclusion:** The system is **consistent and has a unique solution**.

2. Parameter-Based Consistency Conditions

Example 3: Finding a Condition for Consistency

Problem: Find the condition under which the system is consistent:

$$3x + 4y + 5z = \alpha, \quad 4x + 5y + 6z = \beta, \quad 5x + 6y + 7z = \gamma$$

Solution:

1. Construct the augmented matrix $[A \mid B]$:

$$[A \mid B] = \left[\begin{array}{ccc|c} 3 & 4 & 5 & \alpha \\ 4 & 5 & 6 & \beta \\ 5 & 6 & 7 & \gamma \end{array} \right]$$

2. Perform row operations to reduce the matrix:

- $R_3 \rightarrow R_3 - R_2 \implies \left[\begin{array}{ccc|c} 3 & 4 & 5 & \alpha \\ 4 & 5 & 6 & \beta \\ 1 & 1 & 1 & \gamma - \beta \end{array} \right]$
- Swap and simplify to eliminate terms, eventually yielding:

$$\sim \left[\begin{array}{ccc|c} 3 & 4 & 5 & \alpha \\ 0 & -1 & -2 & 3\beta - 4\alpha \\ 0 & 0 & 0 & \gamma - 2\beta + \alpha \end{array} \right]$$

3. **Analysis:** For the system to be consistent, the rank of A must equal the rank of $[A \mid B]$. Since the left side has a zero row, $\rho(A) = 2$. Therefore, the augmented part must also evaluate to zero to prevent rank mismatch.

4. **Conclusion:** The system is consistent only if $\alpha - 2\beta + \gamma = 0$.

Example 4: Investigating Parameters for Unique, None, or Infinite Solutions

Problem: Investigate the values of λ and μ so that the system has:

1. A unique solution
2. No solution
3. An infinite number of solutions

Equations:

$$2x + 3y + 5z = 9, \quad 7x + 3y - 2z = 8, \quad 2x + 3y + \lambda z = \mu$$

Solution:

1. Write the augmented matrix and reduce it using elementary row operations:

$$[A \mid B] = \left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 2 & 3 & \lambda & \mu \end{array} \right]$$

2. After applying row reductions ($R_1 \rightarrow R_1/2$, followed by elimination), the matrix reaches the form:

$$\sim \left[\begin{array}{ccc|c} 1 & 3/2 & 5/2 & 9/2 \\ 0 & -15/2 & -39/2 & -47/2 \\ 0 & 0 & \lambda - 5 & \mu - 9 \end{array} \right]$$

3. Evaluating the Cases:

- **Unique Solution:** Requires $\rho(A) = \rho(A \mid B) = 3$. This happens when $\lambda - 5 \neq 0$, meaning $\lambda \neq 5$ and μ can be any real number.
- **No Solution:** Requires $\rho(A) \neq \rho(A \mid B)$. This occurs if $\lambda = 5$ while $\mu \neq 9$ (creating a row like $[0 \ 0 \ 0 \mid \text{non-zero}]$).
- **Infinite Solutions:** Requires $\rho(A) = \rho(A \mid B) < 3$. This occurs when $\lambda = 5$ and $\mu = 9$ simultaneously, reducing the rank to 2 with 3 unknowns.

3. Homogeneous Systems of Linear Equations

When $B = 0$ ($AX = 0$):

- **Trivial Solution:** If $\rho(A) = n$ (number of unknowns), the system has only the zero solution ($x = y = z = 0$).
- **Infinite Solutions:** If $\rho(A) < n$, the system has $n - r$ linearly independent solutions, resulting in an **infinite number of non-zero solutions**.

Example 5: Solving a Homogeneous System

Problem: Solve the system:

$$x + 2y + 3z = 0, \quad 3x + 4y + 4z = 0, \quad 7x + 10y + 12z = 0$$

Solution:

1. Write the coefficient matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 4 \\ 7 & 10 & 12 \end{bmatrix}$$

2. Apply row transformations ($R_2 \rightarrow R_2 - 3R_1$, $R_3 \rightarrow R_3 - 7R_1$):

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -5 \\ 0 & -4 & -9 \end{bmatrix}$$

3. Apply $R_3 \rightarrow R_3 - 2R_2$:

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

4. **Analysis:** The rank of the coefficient matrix is $\rho(A) = 3$, which matches the number of unknowns ($n = 3$).

5. **Conclusion:** The system has only the **trivial zero solution**: $x = 0, y = 0, z = 0$.